Gregory W. Kyro, PhD

Senior Scientist, AI Research · Lila Sciences
PhD, Computational Biophysical Chemistry · Yale University

gregorykyro.com

gregorykyro@gmail.com · gkyro@lila.ai · gregory.kyro@aya.yale.edu
LinkedIn · Google Scholar · GitHub

Overview

I am a Senior Scientist on the AI Research Team at Lila Sciences, where I develop and lead technical initiatives at the intersection of LLM post-training and experimental science—enabling Lila's central AI agent to learn from proprietary laboratory data, experimental workflows, and feedback from the physical world. Selected work includes:

- **Architected Lila's >10-trillion-token data generation engine that produced the foundation of Lila's post-training corpus.** Created a scalable system for converting experimental data into natural-language scientific tasks with programmatic verification, generating more than 10 trillion high-quality scientific tokens that now form the foundation of Lila's post-training corpus. The system is the subject of a patent application on which I am the lead inventor.
- **Orchestrated Lila's first lab-in-the-loop AI recursive self-improvement cycle.** Spearheaded the cross-functional effort that enabled Lila's first lab-in-the-loop scientific self-improvement cycle, in which the company's central AI agent proposed experiments, received measurements from an automated laboratory workflow, and used the resulting feedback to iteratively refine subsequent experimental designs—establishing Lila's first closed-loop implementation of scientific self-play against physical reality.
- **Built Lila's company-wide framework for translating domain-specific data, workflows, and intelligence into reinforcement learning environments.** Developed Lila's shared framework for transforming domain-specific proprietary data, real laboratory workflows, and scientific objectives into reinforcement learning environments used to train the company's central model. I now lead its implementation throughout cross-organizational scientific AI projects to ultimately drive distributed domain-specific efforts toward coherent programs that advance the scientific capabilities of Lila's AI agent.
- **Led post-training research for scientific reasoning and generalization.** Drove multiple research directions in LLM post-training to improve scientific reasoning and cross-domain generalization, contributing to the design of Lila's post-training program.

Prior to joining Lila, I earned my PhD in Computational Biophysical Chemistry from Yale University in three years as an NSF Fellow, developing AI models that advanced the state of the art in drug discovery and were implemented in collaboration with leading companies including Pfizer, NVIDIA, Novartis, Moderna, Merck Group, and others. Selected research includes:

- **HAC-Net**: AI model for predicting protein-ligand binding affinity—used to identify therapeutic candidates for cancer, diabetes, multiple sclerosis, and drug-resistant infections; and to enable a prostate cancer imaging platform.
- **CardioGenAI**: generative AI framework for re-engineering drugs for reduced hERG-related cardiotoxicity—implemented at Pfizer R&D; evolved into SafeSynthAI—first-place winner of the Blavatnik Fund for Innovation at Yale Ventures, seeding a spinout now in development.
- **T-ALPHA**: successor to HAC-Net; AI model for predicting protein-ligand binding affinity—used to identify therapeutic candidates for thoracic aortic aneurysm.
- **ChemSpaceAL**: active learning method for target-conditioned molecular generation—used to design small-molecule binders to CRISPR-Cas9 to enhance its specificity for target DNA sequences.
- **Electrostatic Eigenvector Centrality**: spectral graph-theoretic method for describing intra-protein information transfer—used to map experimentally validated allosteric mechanisms in CRISPR-Cas9, IGPS, and MIF-2.

Concurrently with my doctoral research, I drove scientific initiatives that drew millions of dollars in government funding, developed technology adopted across the pharmaceutical industry, founded and led scientific organizations, advanced frontier scientific-AI efforts across leading technology and life-science companies, and earned broad external recognition for scientific achievement and leadership—winning prestigious competitions, receiving distinguished awards, and being featured across major scientific media outlets. Selected impact includes:

- **Led the foundational science that catalyzed a $10M state investment in a quantum initiative of which I was a founding scientist.** Served as one of the founding scientists of QuantumCT—a Yale-affiliated initiative aimed at accelerating quantum research partnerships between public and private sectors—where I led the foundational scientific efforts, established collaborations with numerous industry partners including Pfizer, NVIDIA, Novartis, and Moderna, and published the initiative's first collaborative paper as first author—efforts that ultimately catalyzed a $10M investment from the Connecticut State government.
- **Co-developed SAGE—an AI-based rapid-synthesis framework that was acquired by Merck Group** and is currently being integrated across multiple teams within the company.
- **Founded and served as President of the Yale University Chapter of the Biophysical Society**, and mentored researchers who subsequently joined xAI, NVIDIA, and the lab of Nobel laureate David Baker.
- **Won first place at highly selective competitions including the Merck Innovation Cup, with awards from leading federal science agencies such as the National Science Foundation and the National Institutes of Health**—garnering coverage across major scientific media outlets including *Yale News*, the *Yale Alumni Magazine*, and the *Biophysical Society*.
- **Published 18 papers in top-tier journals including *Chemical Reviews*, *Nature Computational Science*, and *Nature Communications***; delivered more than 30 talks, lectures, panels, and conference presentations; and released multiple open-source scientific software packages.
- **Advanced frontier scientific-AI efforts across OpenAI, NVIDIA, and Moderna**—contributing expert scientific data annotation to an OpenAI LLM post-training effort through Scale AI, designing quantum machine learning methods in collaboration with NVIDIA and Moderna, and developing a PROTAC virtual screening pipeline at OpenEye, Cadence Molecular Sciences.
- **Built a network of >10 research collaborations across leading pharma companies and top academic labs—**spanning Drug Safety & Toxicology at Pfizer; Quantum Algorithm Engineering at NVIDIA; Computational Protein Design & Modeling at Moderna; Preclinical Safety at Novartis; Computational Antibody & Protein Engineering at Boehringer Ingelheim; the Schwartz (Cardiology), Hafler (Neuroimmunology), and Kaminski (Pulmonology) Labs at the Yale School of Medicine; the Lisi Lab (Molecular Biology & Biochemistry) at Brown University; the Palermo Lab (Bioengineering) at UC Riverside; and the Loria Lab (Chemistry & Biophysics) at Yale University.

Contents

Education
Research Experience
Innovations Acquired by Pharma
Professional & Leadership Experience
Awards, Scholarships, & Honors
Selected Publications
Selected Presentations & Lectures
Teaching Experience
Scientific Peer Review Contributions
Media Coverage
Selected Research Mentees
Selected Open-Source Software
Apps
Selected Professional Development
Professional Memberships
Selected PhD Research Collaborations
Technical Skills
Foundational Skills
Selected Personal Achievements
Volunteer Activities
Selected Coursework
References

Education

Yale University

PhD, Computational Biophysical Chemistry 05/23 – 02/25

- **Advisor:** Prof. Victor S. Batista
- **Dissertation:** Deep Learning Methods for Modeling Protein–Small Molecule Interactions with Applications to Early-Stage Drug Discovery

MS, Computational Biophysical Chemistry 09/21 – 12/23

- **GPA:** 4.0 / 4.0

SUNY Binghamton

BS, Chemistry 09/16 – 05/21

- **Honors Thesis:** Photophysics of Binuclear Rhenium(I) Tricarbonyl Complexes
- **Major GPA:** 4.0 / 4.0
- **Minors:** Mathematics, Biology

Research Experience

Lila Sciences

Senior Scientist, AI Research 09/26 – Present

Scientist II, AI Research 02/26 – 09/26

Scientist I, AI Research 03/25 – 02/26

- Developed and led technical initiatives at the intersection of LLM post-training and experimental science—enabling Lila's central AI agent to learn from proprietary laboratory data, experimental workflows, and feedback from the physical world.

Laboratory of Prof. Victor S. Batista, Yale University 09/21 – 02/25

National Science Foundation Graduate Research Fellow, PhD Candidate

- Developed numerous state-of-the-art deep learning frameworks for molecular design and protein-ligand binding prediction, spanning use cases in oncology, infectious disease, gene editing, and cardiotoxicity, with multiple methods deployed at major pharmaceutical companies such as Pfizer and Novartis.

Pfizer Research & Development 05/24 – 12/24

Researcher, Computational Safety Sciences (Contractor)

- Developed a generative deep learning framework for re-engineering drug candidates for reduced hERG liability while preserving their primary pharmacology, and applied it to internal drug development programs dealing with hERG-related challenges.

QuantumCT (via Yale University) 05/23 – 05/24

Founding Scientist

- Led the foundational scientific efforts, established collaborations with numerous industry partners including Pfizer, NVIDIA, Novartis, and Moderna, and published the initiative's first collaborative paper as first author—efforts that ultimately catalyzed a $10M investment from the Connecticut State government.

OpenAI (via Scale AI) 03/24 – 06/24

Scientific Data Annotator (Contractor)

- Contributed to the fine-tuning of OpenAI's large language models by crafting scientifically rigorous Q&A-based prompts pertaining to mathematics, physics, and chemistry to enhance the models' scientific domain expertise.

OpenEye, Cadence Molecular Sciences 05/23 – 08/23

Scientific Software Developer (Intern)

- Led the development of a deep learning model for classifying protein-protein interaction interfaces which was integrated into the company's virtual screening pipeline for PROTACs.

NASA (via PreScouter)

06/21 – 09/21

Science & Technology Consultant

- Consulted for NASA on cutting-edge advancements pertaining to energy storage, privacy-preserving networks, image detection, aerial surveillance, and aerodynamic levitation.

Laboratory of Prof. Alistair J. Lees, SUNY Binghamton

08/17 – 06/21

Undergraduate Researcher

- Revealed insights into excited-state mechanisms in binuclear rhenium(I)-based organometallic systems using computational methods (quantum chemistry calculations) and experimental spectroscopy (NMR, UV-Vis, fluorescence, and IR spectroscopy), ultimately leading to anion sensor-based applications.

Innovations Acquired by Pharma

SAGE — a platform that incorporates AI, cutting-edge reactions, and automated synthesis to reduce attrition in early-stage drug discovery. **Acquired by** Merck Group (08/24).

Professional & Leadership Experience

Founder & President | Yale University Chapter of the Biophysical Society	01/24 – Present
Reviewing Editor | Springer Nature	09/25 – Present
Biophysics Research Seminar Organizer | Yale University	08/22 – 05/24
Founder, Mentor & Tutor | Illuminating Minds Academy	09/23 – 01/24
Scientific Blog Writer | Biophysical Society	02/23 – 04/23
STEM Tutor | Transformation Tutoring	12/21 – 12/22
Research Ambassador | Undergraduate Research Center at SUNY Binghamton	08/19 – 06/21
Network and Computer Systems Intern | Rapid Access Communications Inc.	11/19 – 02/20

Awards, Scholarships, & Honors

Industry

- First Place, Merck Innovation Cup \| Merck Group — 07/24

Government

- Graduate Research Fellowship \| National Science Foundation — 03/23
- Award No. 5T32GM008283-35 \| National Institutes of Health — 09/22

Academia

- Stanley K. Madan Award in Inorganic Chemistry \| SUNY Binghamton — 05/21
- Honors Thesis in Chemistry \| SUNY Binghamton — 05/21
- Summer Scholars Program Award \| SUNY Binghamton — 06/21
- Undergraduate Research Award, Spring 2020 \| SUNY Binghamton — 01/20
- Undergraduate Research Award, Fall 2019 \| SUNY Binghamton — 09/19
- Undergraduate Research Award, Spring 2019 \| SUNY Binghamton — 01/19
- Undergraduate Research Award, Fall 2018 \| SUNY Binghamton — 09/18
- Undergraduate Research Award, Spring 2018 \| SUNY Binghamton — 01/18
- Undergraduate Research Award, Fall 2017 \| SUNY Binghamton — 09/17

Selected Publications

[18] Shee, Y; Smaldone, AM; **Kyro, GW**; Batista, VS. "FragmentRetro: A Quadratic Retrosynthetic Method Based on Fragmentation Algorithms." *Journal of Chemical Theory and Computation*. 2026. DOI: 10.1021/acs.jctc.5c01632

[17] **Kyro, GW**; Smaldone, AM; Shee, Y; Xu, T; Batista, VS. "T-ALPHA: A Hierarchical Transformer-Based Deep Neural Network for Protein–Ligand Binding Affinity Prediction with Uncertainty-Aware Self-Learning for Protein-Specific Alignment." *Journal of Chemical Information and Modeling*. 2025, 65, 5. DOI: 10.1021/acs.jcim.4c02332

[16] **Kyro, GW**; Martin, MT; Watt, ED; Batista, VS. "CardioGenAI: A Machine Learning-based Framework for Re-engineering Drugs for Reduced hERG Liability." *Journal of Cheminformatics*. 2025, 17, 30. DOI: 10.1186/s13321-025-00976-8

[15] **Kyro, GW**; Qiu, T; Batista, VS. "A Model-Centric Review of Deep Learning for Protein Design." *arXiv*. 2025. DOI: 10.48550/arXiv.2502.19173

[14] Smaldone, AM; Shee, Y; **Kyro, GW**; Farag, MH; Chandani, Z; Kysoeva, E; Batista, VS. "A Hybrid Transformer Architecture with a Quantized Self-Attention Mechanism Applied to Molecular Generation." *Journal of Chemical Theory and Computation*. 2025, 21, 10. DOI: 10.1021/acs.jctc.5c00331

[13] Smaldone, AM; **Kyro, GW**; Shee, Y; Batista, VS. "Quantum Machine Learning in Drug Discovery: Applications in Academia and Pharmaceutical Industries." *Chemical Reviews*. 2025. DOI: 10.1021/acs.chemrev.4c00678

[12] **Kyro, GW**; Morgunov, A; Brent, RI; Batista, VS. "ChemSpaceAL: An Efficient Active Learning Methodology Applied to Protein-Specific Molecular Generation." *Journal of Chemical Information and Modeling*. 2024, 64, 3, 653-665. DOI: 10.1021/acs.jcim.3c01456

[11] **Kyro, GW**; Batista, VS. "Hierarchical Cross-Scale Transformer Architecture for Bottom-Up Reasoning." *TechRxiv*. 2024. DOI: 10.36227/techrxiv.173121338.85984992/v1

[10] **Kyro, GW**; Brent, RI; Batista, VS. "HAC-Net: A Hybrid Attention-based Convolutional Neural Network for Highly Accurate Protein-Ligand Binding Affinity Prediction." *Journal of Chemical Information and Modeling*. 2023, 63, 7, 1947-1960. DOI: 10.1021/acs.jcim.3c00251

[9] Smaldone, AM; **Kyro, GW**; Batista, VS. "Quantum Convolutional Neural Networks for Multi-Channel Supervised Learning." *Quantum Machine Intelligence*. 2023, 5, 41. DOI: 10.1007/s42484-023-00130-3

[8] Maschietto, F; Allen, B; **Kyro, GW**; Batista, VS. "MDiGest: A Python Package for Describing Allostery from Molecular Dynamics Simulations." *Journal of Chemical Physics*. 2023, 158, 215103. DOI: 10.1063/5.0140453

[7] Yang, KR; **Kyro, GW**; Batista, VS. "The Landscape of Computational Approaches for Artificial Photosynthesis." *Nature Computational Science*. 2023, 3, 504-513. DOI: 10.1038/s43588-023-00450-1

[6] Maschietto, F; Morzan, U; Tofoleanu, F; Gheereart, A; Chaudhuri, A; **Kyro, GW**; Nekrasov, P; Brooks, B; Loria, JP; Rivalta, I; Batista, VS. "Turning Up the Heat Mimics Allosteric Signaling in Imidazole-Glycerol Phosphate Synthase." *Nature Communications*. 2023, 14, 2239. DOI: 10.1038/s41467-023-37956-1

[5] Chen, E; Widjaja, V; **Kyro, GW**; Allen, B; Das, P; Bhandari, V; Lolis, EJ; Batista, VS; Lisi, GP. "Mapping N- to C-terminal Allosteric Coupling Through Disruption of the Putative CD74 Activation Site in D-Dopachrome Tautomerase." *Journal of Biological Chemistry*. 2023, 299, 6, 104729. DOI: 10.1016/j.jbc.2023.104729

[4] Maschietto, F; **Kyro, GW**; Allen, B; Batista, VS. "Electrostatic Networks for Characterization of Allosteric Pathways in Cas9 Apo, RNA- and DNA-Bound Forms." *Biophysical Journal*. 2023, 122 (3). DOI: 10.1016/j.bpj.2022.11.389

[3] Wang, J; Arantes, PR; Ahsan, M; Sinha, S; **Kyro, GW**; Maschietto, F; Allen, B; Skeens, E; Lisi, GP; Batista, VS; Palermo, G. "Twisting and Swiveling Domain Motions in Cas9 to Recognize Target DNA Duplexes, Make Double-Strand Breaks, and Release Cleaved Duplexes." *Frontiers in Molecular Biosciences*. 2023, 9. DOI: 10.3389/fmolb.2022.1072733

[2] Wang, J; Skeens, E; Arantes, P; Maschietto, F; Allen, B; **Kyro, GW**; Lisi, GP; Palermo, G; Batista, VS. "Structural Basis for Reduced Dynamics of Three Engineered HNH Endonuclease Lys-to-Ala Mutants for the CRISPR-Associated 9 (CRISPR/Cas9) Enzyme." *Biochemistry*. 2022, 61 (9), 785-794. DOI: 10.1021/acs.biochem.2c00127

[1] **Kyro, GW**; Lees, AJ. "Photophysics of Rhenium(I) Polypyridyl-based Complexes and Their Employment as Highly Sensitive Anion Sensors." 2021. DOI: 10.13140/RG.2.2.29980.56962

Selected Presentations & Lectures

[33] Industry Trends in AI for Scientific Discovery. Invited Interview at *PhD Pathways Speaker Series, Yale University* (2025).

[32] Artificial Intelligence for Preclinical Drug Safety. Invited Round-Table Discussion at *Novartis Institutes for Biomedical Research* (2025).

[31] Modeling Protein-Ligand Interactions and Generative AI for Lead Optimization. Invited Round-Table Discussion at *Novartis Institutes for Biomedical Research* (2025).

[30] Quantum-Classical Machine Learning Methods for Optimizing Drug Toxicity. Invited Research Talk at *QuantumCT Industry Collaboration Forum, Yale Ventures* (2025).

[29] Exploring How Quantum Computing Will Impact Pharmaceutical Research. Invited Technology Panel Discussion at *QuantumCT Industry Collaboration Forum, Yale Ventures* (2025).

[28] Current State-of-the-Art Deep Learning Models for Protein Design. Invited Research Talk at *Biophysical Chemistry Seminar, Yale University* (2025).

[27] Introduction to Deep Learning for Biochemistry. Invited Guest Lecture at *CHEM 584: Machine Learning and Quantum Computing, Yale University* (2025).

[26] Transformers for Modeling Protein-Ligand Interactions. Poster Presentation at *Chemical Research Symposium, Yale University* (2025).

[25] A Hybrid Quantum-Classical Transformer Architecture with a Quantized Self-Attention Mechanism Applied to Molecular Generation. Poster Presentation at *Chemical Research Symposium, Yale University* (2025).

[24] Applications of Deep Learning in Cardiovascular and Safety Domains. Invited Round-Table Discussion at *Novartis Institutes for Biomedical Research* (2025).

[23] Machine Learning for Modeling Cardiac Ion Channels. Invited Research Talk at *Scientific Seminar, Novartis Institutes for Biomedical Research* (2025).

[22] T-ALPHA: A Hierarchical Transformer-Based Deep Neural Network for Protein–Ligand Binding Affinity Prediction with Uncertainty-Aware Self-Learning for Protein-Specific Alignment. Invited Research Talk at *10th Annual Biophysics and Structural Biology Research Symposium, Yale University* (2025).

[21] Assessment of Different Machine Learning Architectures for hERG Activity Prediction. Poster Presentation at *Computational Medicinal Chemistry School, Novartis Institutes for Biomedical Research* (2024).

[20] Some of the Problems with Public Datasets for Protein-Ligand Binding Affinity Prediction. Invited Research Talk at *Computational Biophysics and Drug Design Meeting, Bernal Institute at the University of Limerick* (2024).

[19] Generative AI Methods for Lead Optimization in Drug Discovery. Poster Presentation at *Chemical Research Symposium, Yale University* (2024).

[18] CardioGenAI: A Machine Learning Framework for Re-engineering Drugs for Reduced hERG Liability. Invited Research Talk at *Innovation Cup Alumni Symposium, Merck Group* (2024).

[17] A Machine Learning Framework for Re-engineering Drugs for Reduced hERG Liability. Invited Research Talk at *Global Discovery Investigative Toxicology and Translational Sciences—Computational Safety Sciences Town Hall, Groton Laboratories at Pfizer Research & Development* (2024).

[16] CardioGenAI: A Machine Learning-based Framework for Re-engineering Drugs for Reduced hERG Liability. Poster Presentation at *Summer Intern Poster Session, Groton Laboratories at Pfizer Research & Development* (2024).

[15] A Generative AI-based Framework for Toxicity Applications in Early-Stage Drug Development. Poster Presentation at *9th Annual Biophysics and Structural Biology Research Symposium, Yale University* (2024).

[14] A Hybrid Quantum-Classical Machine Learning Framework for Drug Toxicity Applications. Invited Research Talk at *QuantumCT Industry Collaboration Forum, Yale West Campus* (2024).

[13] Generative Machine Learning and Active Learning Methods for Hit Identification in Drug Discovery. Poster Presentation at *Sterling Chemistry Laboratory 101st Anniversary Symposium, Yale University* (2024).

[12] CardioGenAI: A Machine Learning-based Framework for Re-engineering Drugs for Reduced hERG Liability. Poster Presentation at *19th Annual Drug Discovery Chemistry Conference* (2024).

[11] ChemSpaceAL: An Efficient Active Learning Methodology Applied to Protein-Specific Molecular Generation. Invited Research Talk at *Annual Biophysical Society Meeting* (2024).

[10] Machine Learning and Statistical Methods for Modulating Protein Function with Small Molecule Inhibitors. Invited Research Talk at *National Institutes of Health Biophysics Seminar, Yale University* (2023).

[9] HAC-Net: A Hybrid Attention-based Convolutional Neural Network for Highly Accurate Protein-Ligand Binding Affinity Prediction. Poster Presentation at *Annual Biophysical Society Meeting* (2023).

[8] Introduction to Deep Learning for Chemistry. Invited Guest Lecture at *CHEM 584: Machine Learning and Quantum Computing, Yale University* (2022).

[7] Photophysics of Binuclear Rhenium(I) Tricarbonyl Complexes and Their Employment as Anion Sensors Through Charge-Mediated Hydrogen Bonding. Poster Presentation at *261st American Chemical Society National Meeting & Exposition* (2021).

[6] Variable Anion Recognition Sites in Phosphorescent Rhenium(I) Polypyridyl-based Sensors. Poster Presentation at *259th American Chemical Society National Meeting & Exposition* (2020).

[5] Photophysics of Polypyridyl-based Rhenium(I) Complexes and Their Employment as Anion Sensors. Poster Presentation at *3rd SUNY Binghamton Conference in Chemistry Research* (2020).

[4] Highly Sensitive Rhenium(I) Sensors for Anions Through Amide Hydrogen Bonding. Poster Presentation at *Undergraduate Research Conference, SUNY Binghamton* (2020).

[3] Amide Protons as Binding Groups in a Polypyridyl-based Rhenium(I) Anion Sensor. Poster Presentation at *257th American Chemical Society National Meeting & Exposition* (2019).

[2] Excited-State Properties of Rhenium(I)-based Anion Sensors. Poster Presentation at *2nd SUNY Binghamton Conference in Chemistry Research* (2019).

[1] Organometallic Complexes as Anion Sensors: A Highly Sensitive Rhenium(I) Complex for Cyanide and Halide Anions. Poster Presentation at *1st SUNY Binghamton Conference in Chemistry Research* (2018).

Teaching Experience

Yale University

Teaching Fellow

- Matrix Methods in Quantum Mechanics 10/23 – 12/23
- Machine Learning & Quantum Computing 03/23 – 06/23
- Advanced General Chemistry II 01/25 – 05/25
- General Chemistry I 08/24 – 12/24

SUNY Binghamton

Teaching Assistant

- Inorganic Chemistry, Fall 2020 09/20 – 12/20
- Chemical Principles I, Fall 2020 09/20 – 12/20
- Inorganic Chemistry, Fall 2019 09/19 – 12/19
- Chemical Principles II, Spring 2019 01/19 – 05/19
- Introduction to Chemistry, Fall 2017 09/17 – 12/17

Scientific Peer Review Contributions

Reviewing Editor

- Springer Nature

Trusted Reviewer

- *Nature Communications* (Impact Factor: 15.7) | Nature Portfolio
- *npj Digital Medicine* (Impact Factor: 15.1) | Nature Portfolio
- *Environmental Science & Technology* (Impact Factor: 11.4) | American Chemical Society
- *Journal of Cheminformatics* (Impact Factor: 8.6) | BioMed Central
- *Engineering Applications of Artificial Intelligence* (Impact Factor: 7.5) | Elsevier
- *EPJ Quantum Technology* (Impact Factor: 5.8) | Springer
- *Journal of Chemical Theory and Computation* (Impact Factor: 5.7) | American Chemical Society
- *Journal of Chemical Information and Modeling* (Impact Factor: 5.6) | American Chemical Society
- *Computational and Structural Biotechnology Journal* (Impact Factor: 4.5) | Elsevier
- *Biotechnology* (Impact Factor: 4.4) | Oxford University Press
- *ACS Synthetic Biology* (Impact Factor: 4.3) | American Chemical Society
- *BMC Bioinformatics* (Impact Factor: 3.9) | BioMed Central
- *Frontiers in Bioinformatics* (Impact Factor: 3.9) | Frontiers
- *Scientific Reports* (Impact Factor: 3.8) | Nature Portfolio
- *Chemical Research in Toxicology* (Impact Factor: 3.8) | American Chemical Society
- *European Journal of Nuclear Medicine & Molecular Imaging Research* (Impact Factor: 3.1) | Springer
- *Journal of Computer-Aided Molecular Design* (Impact Factor: 3.0) | Springer
- *Journal of Molecular Modeling* (Impact Factor: 2.1) | Springer
- *npj Drug Discovery* (Impact Factor: Pending) | Nature Portfolio
- *npj Artificial Intelligence* (Impact Factor: Pending) | Nature Portfolio

Media Coverage

- Featured by the Biophysical Society as the founder and president of the Yale University Chapter to articulate a first-principles perspective on biophysics and intellectual rigor.
- Featured by Merck Group for winning first place at the 2024 Merck Innovation Cup.
- Featured in the *Yale Alumni Magazine* for sharing insights on biochemistry research.
- Research featured in the NVIDIA company blog for simulating quantum processors in parallel on GPUs for drug discovery.
- Featured in *Yale News* for becoming a National Science Foundation fellow.
- Featured by the Yale University Chapter of the Biophysical Society for contributing to quantum machine learning in drug discovery.
- Featured in the *Biophysical Society Blog* for sharing opinions about work presented at the 2023 Biophysical Society Annual Meeting.

Selected Research Mentees

- **Tsach Mackey** — Technical Staff, xAI
- **Rafael I. Brent** — PhD candidate, David Baker Lab
- **Gavin Wong** — AI Researcher, NVIDIA
- **Akilan Sankaran** — AI Research Intern, Lila Sciences

Selected Open-Source Software

- **T-ALPHA** — a hierarchical transformer-based deep neural network for protein–ligand binding affinity prediction with uncertainty-aware self-learning for protein-specific alignment | github.com/gregory-kyro/T-ALPHA
- **CardioGenAI** — a machine learning-based framework for re-engineering drugs for reduced hERG liability | github.com/gregory-kyro/CardioGenAI
- **ChemSpaceAL** — an efficient active learning methodology applied to protein-specific molecular generation | github.com/gregory-kyro/ChemSpaceAL
- **HAC-Net** — a hybrid attention-based convolutional neural network for highly accurate protein–ligand binding affinity prediction | github.com/gregory-kyro/HAC-Net

Apps

- **BibleVerse** — a modern Bible interface with latent-space verse and passage retrieval, scripture-augmented LLMs designed for breadth- and depth-first knowledge expansion, and a reader with chapter-conditioned LLM chat and annotation | bibleverse-ai.com

Selected Professional Development

Massachusetts Institute of Technology, edX

- Machine Learning With Python 06/22
- Biochemistry: Biomolecules, Methods & Mechanisms 05/22
- Computational Thinking Using Python 07/21

Harvard University, edX

- Principles of Biochemistry 05/22
- Using Python for Research 07/21

Professional Memberships

- American Chemical Society
- Biophysical Society
- Yale Alumni Association
- Boston Biotech
- QuantumCT
- OpenLabs at Yale

Selected PhD Research Collaborations

Industry Collaborations

- Pfizer — Drug Safety and Toxicology
- NVIDIA — Quantum Algorithm Engineering
- Moderna — Computational Protein Design and Modeling
- Novartis — Preclinical Safety
- Boehringer Ingelheim — Computational Antibody and Protein Engineering

Academic Collaborations

- Yale School of Medicine — Schwartz Lab (Cardiology); Hafler Lab (Neuroimmunology); Kaminski Lab (Pulmonology)
- Brown University — Lisi Lab (Molecular Biology & Biochemistry)
- University of California, Riverside — Palermo Lab (Bioengineering)
- Yale University — Loria Lab (Chemistry & Biophysics)

Technical Skills

- **Deep Learning Architectures:** transformers, diffusion models, GANs, autoencoders, RNNs, LSTMs, GRUs, CNNs, GNNs, LLMs, and more
- **Machine Learning Architectures:** linear regression, logistic regression, decision trees, random forests, support vector machines, gradient boosting machines, and more
- **Machine Learning Techniques:** active learning, reinforcement learning, transfer learning, feature engineering, dimensionality reduction, regularization, hyperparameter optimization, ensemble methods, cross-validation, clustering, data preprocessing, and more
- **Python Software Packages:** PyTorch, TensorFlow, Scikit-Learn, XGBoost, PyG, RDKit, BioPython, MDAnalysis, PyMOL, Open Babel, ProLIF, OpenMM, PDBFixer, Optuna, SciPy, NumPy, Pandas, Matplotlib, Seaborn, and more
- **Cheminformatics:** molecular property prediction, molecular interaction analysis, molecular modeling, virtual screening, ligand-based drug design, structure-based drug design, molecular dynamics simulation analysis, molecular feature representations, quantitative structure-activity relationship modeling, molecular similarity analysis, conformational analysis, protein visualization, molecular mechanical calculations, quantum chemistry calculations, and more
- **Data Analysis:** statistical and mathematical modeling, time series analysis, visualization, network analysis, optimization techniques, and more
- **Quantum Computing:** quantum machine learning, quantum circuit construction, quantum algorithms, and more

Foundational Skills

- **Problem Solving & Critical Thinking:** first-principles reasoning, creativity skills, optimization, and more
- **Leadership & Project Management:** idea generation, time management, multitasking, strategic thinking, mentorship, and more
- **Communication & Collaboration:** public speaking, technical writing, interpersonal skills, and more
- **Adaptability & Execution:** extreme discipline, versatility, curiosity, sangfroid, growth mindset, and more

Selected Personal Achievements

- Have done a high-intensity workout at least once every day since 2010 (no exceptions).
- 2944 puzzle rating (99.9th percentile) and 1999 bullet rating (99.8th percentile) on chess.com.
- Weight-lifting stats: 335 lb bench press, 405 lb squat, 495 lb deadlift (at ~180 lb bodyweight).
- Scored a perfect 28/28 in the New York State School Music Association guitar competition at level 4 at age 8.
- Five-sport athlete in high school (baseball, wrestling, basketball, track, football).

Volunteer Activities

Scientific Speaker | Yale Pathways to Science	12/23 – Present
Outreach Volunteer | American Chemical Society	06/19 – 06/20
Medical Volunteer | Long Island Jewish Medical Center	06/17 – 01/19
Patient Care Volunteer | Ronald McDonald House Charities	06/17 – 12/18
Gift of Sight Volunteer | Luxottica Group S.p.A.	05/16 – 08/16
Special Education Volunteer | Merillon Little League Baseball	06/15 – 08/15

Selected Coursework

Yale University

- **Machine Learning & Computational Modeling:** Machine Learning & Quantum Computing, Computational Chemistry, Computer Science & The Modern Intellectual Agenda
- **Quantum Physics & Statistics:** Advanced Quantum Mechanics, Statistical Mechanics I and II, Quantum Mechanics I and II
- **Biophysics:** Biochemical Rates & Mechanisms I and II, Quantitative Biochemical Imaging, Biophysical Optical Spectroscopy

SUNY Binghamton

- **Quantum Systems & Chemical Dynamics:** Quantum Chemistry, Physical Chemistry, Molecular Photochemistry
- **Molecular Biophysics & Biochemistry:** Biophysical Chemistry, Molecular Biology, Molecular Genetics
- **Chemistry Principles:** Intermediate Inorganic Chemistry, Chemical Principles I and II, Transition Metal Chemistry, Organic Chemistry I and II, Analytical Chemistry
- **Mathematical, Physical & Data Science Principles:** Infinite Series, Integration Techniques & Application, Integral Calculus, Differential Calculus, General Physics I and II, Biostatistics
- **Biological Systems:** Organismal Biology, Evolutionary Biology, Human Biology & Health

References

Victor S. Batista, Yale University

- John Gamble Kirkwood Professor of Chemistry · PhD Advisor · victor.batista@yale.edu

Matthew T. Martin, Pfizer R&D

- Executive Director, Global Head of Computational Safety Sciences · Research Supervisor · matthew.martin@pfizer.com

Vishal Vaidya, Pfizer R&D

- Vice President, Chief Toxicology Scientist · Research Collaborator and Career Mentor · vishal.vaidya@pfizer.com

Alistair J. Lees, SUNY Binghamton

- Professor of Chemistry · Undergraduate Research Advisor · alees@binghamton.edu

John R. Swierk, SUNY Binghamton

- Professor of Chemistry · Undergraduate Academic Advisor and Honors Thesis Committee Member · jswierk@binghamton.edu

J. Patrick Loria, Yale University

- Professor of Chemistry & Biophysics, Director of Graduate Studies · Research Collaborator and Thesis Committee Member · patrick.loria@yale.edu

Tianyu Zhu, Yale University

- Professor of Chemistry · Research Collaborator and Thesis Committee Member · tianyu.zhu@yale.edu